

Homework 1 Report

Dataset Information

Dataset: <https://archive.ics.uci.edu/dataset/2/adult/adult.data>

Instances: 48842

Features: 14

Goal: To predict whether income is greater than 50K

Perturb the data

Data preprocessing:

1. Remove the instances with NaN and reset the index of the remaining instances.
(48842 -> 32561)
2. Use `LabelEncoder()` to normalize the feature like "sex" and "incomes"
 - "sex": Female -> 0, Male -> 1
 - "incomes": <=50K -> 0, >50K -> 1

K-anonymity:

1. Choose "age", "education-num", "hours-per-week" as quasi-identifier.
2. `mondrian_partition()` function slices the data into as many groups as possible based on the distribution of quasi-identifiers, while ensuring that there are at least k pieces of data in each group.
 - Check if the number of quasi-identifiers in the group is less than $2 \times k$, it means that the group cannot be cut anymore, so we will return to the group directly.
 - Find the field with the largest max-min data range among all quasi-identifiers, and use it as the basis for cut.
 - Split the data into left and right halves according to the median of the field.
 - Recursive partitioning: call `mondrian_partition()` for each half of the partitioned data until all subgroups meet the K-anonymity requirement.
3. `generalize_partition()` function makes all records in a group look the same on quasi-identifiers, thus achieving K-anonymity.
 - Group-by-group processing: For each partitioned group, take the minimum and maximum values of each quasi-identifier field and apply the generalization format, e.g. age = [25-35].
 - The quasi-identifiers for each piece of data are replaced with the

generalized values for that group.

- Finally, use `pd.concat()` to combine all the generalized data into one dataframe.

Machine Learning Model

Model 1: Support Vector Machine (SVM)

Train-test split: 80% training data, 20% testing data.

Effectiveness measure: Misclassification Rate, Accuracy, Precision, Recall, and AUC

SVM Evaluation

	Original	K=2	K=5	K=10
Misclassification Rate	0.1905	0.2412	0.2412	0.2412
Accuracy	0.8095	0.7588	0.7588	0.7588
Precision	0.7007	0	0	0
Recall	0.3666	0	0	0
AUC	0.7994	0.5386	0.5888	0.5414

Confusion matrix:

TP (True Positive): Predicted to be >50K and actually >50K.

FP (False Positive): Predicted to be >50K, but was actually <=50K.

TN (True Negative): Predicted to be <=50K, and actually <=50K.

FN (False Negative): Predicted 0 <=50K, but actually >50K.

Original: TN: 4696, FP: 246, FN: 995, TP: 576

K=2: TN: 4942, FP: 0, FN: 1571, TP: 0

K=5: TN: 4942, FP: 0, FN: 1571, TP: 0

K=10: TN: 4942, FP: 0, FN: 1571, TP: 0

Observation:

From the above results, it can be seen that k-anonymization has a significant negative impact on the performance of the SVM model.

As the value of k increases, the anonymization of the dataset deepens, leading to a significant decrease in the predictive ability of the model, the model fails to effectively distinguish between positive and negative categories, and even fails to predict any positive samples.

Model 2: RandomForest

Train-test split: 80% training data, 20% testing data.

Effectiveness measure: Misclassification Rate, Accuracy, Precision, Recall, and AUC

RandomForest Evaluation

	Original	K=2	K=5	K=10
Misclassification Rate	0.2173	0.2472	0.246	0.2444
Accuracy	0.7827	0.7528	0.754	0.7556
Precision	0.5661	0.2	0.254	0.2857
Recall	0.4252	0.0083	0.0102	0.0089
AUC	0.793	0.6057	0.6089	0.6103

Confusion matrix:

Original: TN: 4430, FP: 512, FN: 903, TP: 668

K=2: TN: 4890, FP: 52, FN: 1558, TP: 13

K=5: TN: 4895, FP: 47, FN: 1555, TP: 16

K=10: TN: 4907, FP: 35, FN: 1557, TP: 14

Observation:

From the above results, it can be seen that k-anonymization also has a significant negative impact on the performance of the RandomForest model.

The performance of Random Forest is also degraded for k=2, k=5, and k=10, but its performance is relatively stable and better than SVM especially in terms of precision and recall.

Although SVM's prediction accuracy on the original dataset is better than Random Forest, if we have to consider the effect after K-anonymization, Random Forest is more suitable for data prediction than SVM.